

PORT SCANNING AND ANALYSIS REPORT

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TOOLS USED:

Nmap

INTRODUCTION

Port scanning is an important network security technique used to identify open ports, active services, and potential vulnerabilities on a system or network. Open ports allow communication between devices and services, but unnecessary or unsecured ports may expose the system to cyber threats.

This assessment was conducted in an authorized local environment using Nmap, a popular network scanning tool. The purpose of the scan was to identify open TCP and UDP ports, analyze active services, recognize potentially high-risk exposed services, and document security recommendations to improve the overall security posture of the system.

OBJECTIVE

The objective of this task was to perform TCP and UDP scans on the local machine, identify active services, recognize high-risk ports, and analyze potential security risks associated with exposed services.

The assessment also aimed to develop practical understanding of:

- Network scanning
- Service detection
- Port analysis
- Basic vulnerability assessment
- Security documentation

SCOPE OF SCAN

The scope of this assessment was limited to the local machine and authorized local network only.

Included in Scope

- TCP port scanning
- UDP port scanning
- Service identification
- Open port analysis
- Risk assessment of exposed services

Excluded from Scope

- External network scanning
- Exploitation testing
- Malware analysis
- Wireless attacks
- Unauthorized systems scanning

The assessment was conducted only on the local system for educational and cybersecurity learning purposes.

TOOLS USED

Tool	Purpose
Nmap	Port scanning and service detection
Command Prompt	Running network scanning commands
Windows Network Tools	IP address identification

COMMANDS USED

Display Local IP Address

ipconfig

Purpose:

Used to identify the local IPv4 address of the system.

TCP SYN Scan

```
nmap -sS 192.168.1.6
```

Purpose:

Performs a TCP SYN scan to identify open TCP ports and active services.

UDP Scan

```
nmap -sU 192.168.1.6
```

Purpose:

Performs a UDP scan to identify active UDP services and open UDP ports.

TCP SCAN RESULTS

The TCP SYN scan identified several open TCP ports associated with Windows networking and system services.

Open TCP Ports

Port	State	Service
135/tcp	Open	MSRPC
139/tcp	Open	NetBIOS-SSN
445/tcp	Open	Microsoft-DS
1042/tcp	Open	Afrog
1043/tcp	Open	BOINC
6646/tcp	Open	Unknown

TCP SCAN ANALYSIS

Port 135 — MSRPC

Port 135 is used by Microsoft Remote Procedure Call services. It allows communication between Windows applications and services. While commonly required for Windows functionality, exposure of this port may increase the risk of remote enumeration and exploitation attempts.

Port 139 — NetBIOS-SSN

Port 139 is associated with NetBIOS Session Service and is commonly used for file and printer sharing within Windows networks. Exposed NetBIOS services may reveal system information to attackers.

Port 445 — Microsoft-DS (SMB)

Port 445 is used for SMB file sharing services. This port is considered high risk because SMB vulnerabilities have been targeted in ransomware attacks such as WannaCry. If SMB is not required, restricting or disabling this service is recommended.

Port 1042 — Afrog

Port 1042 was identified with the Afrog service. Unknown or uncommon services should be monitored carefully to ensure they are legitimate and secure.

Port 1043 — BOINC

Port 1043 is associated with BOINC services. BOINC is commonly used for distributed computing tasks. Unnecessary exposure to such services may increase the attack surface.

Port 6646 — Unknown

Port 6646 was identified as an unknown service. Unknown open ports should be investigated further to verify legitimacy and ensure they are not associated with unauthorized applications.

UDP SCAN RESULTS

The UDP scan was performed to identify active UDP-based services on the local machine.

UDP Scan Observation

The scan result indicated:

- All 1000 scanned UDP ports were in ignored states
- Ports were identified as open|filtered with no response

This means the system did not respond to UDP probes, which is common because many UDP services do not reply unless specifically configured.

UDP SCAN ANALYSIS

No critical UDP services were clearly identified during the assessment. The absence of visible UDP responses reduced the likelihood of immediate UDP-based vulnerabilities. However, filtered UDP ports may still require monitoring depending on network configuration.

HIGH-RISK PORTS ANALYSIS

Port	Service	Risk Level	Security Concern
445	Microsoft-DS (SMB)	High	Vulnerable to ransomware and file-sharing attacks
139	NetBIOS-SSN	Medium	May expose shared resources and system information
135	MSRPC	Medium	Potential remote procedure exploitation
6646	Unknown	Medium	Unknown service exposure
1042	Afrog	Low-Medium	Uncommon service exposure
1043	BOINC	Low-Medium	Distributed computing service exposure

DOCUMENT FINDINGS

The assessment identified multiple active TCP services running on the local machine. Windows networking-related services such as MSRPC, NetBIOS, and SMB were found open and accessible locally.

Port 445 (SMB) was identified as the highest-risk service because SMB vulnerabilities are commonly targeted in ransomware and unauthorized file-sharing attacks.

An unknown service running on Port 6646 was also identified, which requires further investigation to verify whether it is legitimate and secure.

The UDP scan did not identify critical UDP vulnerabilities. Most UDP ports were classified as open|filtered because no response was received during the scan.

The system demonstrated moderate security posture, but exposure of multiple services increases the potential attack surface if left unmonitored.

RECOMMENDATIONS

- Disable unnecessary services and close unused ports
- Restrict SMB access if file sharing is not required
- Investigate unknown services such as Port 6646
- Keep Windows Firewall enabled
- Perform regular vulnerability scans
- Keep the operating system and software updated
- Monitor network traffic for suspicious activity

CONCLUSION

The port scanning assessment successfully identified multiple open TCP ports and active services running on the local machine. The most significant finding was the exposure of SMB services on Port 445, which may increase the risk of ransomware and network-based attacks if improperly configured.

Although no major UDP vulnerabilities were detected, the presence of multiple exposed services highlights the importance of continuous monitoring and proper firewall configuration.

Overall, the assessment provided valuable practical experience in network security analysis, port scanning, and risk identification using Nmap. Implementing the recommended security controls would further improve the security posture of the system.

```
C:\Users\prath>nmap -sU 192.168.1.6
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-19 12:09 +0530
Nmap scan report for ASUS (192.168.1.6)
Host is up.
All 1000 scanned ports on ASUS (192.168.1.6) are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 207.64 seconds

C:\Users\prath>nmap -sS 192.168.1.6
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-19 12:06 +0530
Nmap scan report for ASUS (192.168.1.6)
Host is up (0.0027s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
1042/tcp  open  afrog
1043/tcp  open  boinc
6646/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 58.28 seconds
```

END OF REPORT